

Footprint a Wireless Network

Execution mode	Authorized, isolated training lab; no public systems were targeted.
Output status	Representative/sanitized lab output recorded for submission. Replace with instructor-required screenshots if a live lab run is performed.

Aim

Analyze a pre-captured wireless packet file to identify SSIDs, BSSIDs, channels, encryption, and associated clients.

Theory

Wireless footprinting is safest when performed offline on a supplied capture. Beacon frames advertise network identity and capabilities; data frames reveal client/AP relationships. No live monitor-mode capture was performed.

Tools and Lab Scope

- Wireshark
- Aircrack-ng airodump-ng offline mode
- sample_capture.pcap supplied for the lab

Procedure and Recorded Execution

Step 1. Open the capture - Load the pcap in Wireshark and verify that it is the course sample.

```
file: sample_capture.pcap
packets: 1,248
mode: offline analysis
```

Step 2. Filter beacon frames - Use the beacon subtype filter to list advertised networks.

```
wlan.fc.type_subtype == 0x08
beacons matched: 37
```

Step 3. Read security fields - Inspect RSN information and WEP/open indicators.

Step 4. Identify clients - Filter on the target BSSID and record client MACs with privacy-safe labels.

```
wlan.addr == 00:11:22:33:44:55
associated stations: 2
```

Step 5. Compare with airodump - Use offline read mode and reconcile the table with Wireshark.

```
airodump-ng --read sample_capture.pcap
APs: 3; clients: 5
```

Wireless footprint output

The offline capture contained three access points and five client stations. The inventory below uses lab labels and is not a live scan.

SSID	BSSID	Channel	Security	Clients
LabNet	00:11:22:33:44:55	6	WPA2-PSK/AES	2
GuestLab	00:11:22:33:44:66	11	Open	1
IoT-Training	00:11:22:33:44:77	1	WPA2-PSK/AES	2

Result

Wireless footprinting completed from the pcap. The exercise demonstrated how beacons and data frames reveal network inventory without transmitting or probing a live wireless environment.

Safety and Submission Notes

- Analyze only captures you are authorized to possess.
- MAC addresses are still identifiers; redact them in public reports.
- Never perform live wireless capture outside an authorized lab.

Viva Questions and Answers

What is a beacon frame?

A periodic management frame advertising an access point and its capabilities.

SSID vs BSSID?

SSID is the network name; BSSID is the AP radio MAC address.

How tell WPA2 vs open?

RSN information indicates WPA2; its absence can indicate an open network, subject to context.

Why identify clients?

Client/AP relationships help defenders understand exposure and investigate suspicious associations.

Legal restriction?

Live wireless monitoring must be authorized by the network owner and comply with local law.